

NATASHQUAN, QC, Canada

WMO: 715130

Lat: 50.190N

Lon: 61.790W

Elev: 39

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-11.2	-27.0	1.2	-14.8	-22.2	1.6	-10.0	35.8	20.4	32.2	20.9	7.7	300	0.770

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	71.9	62.9	69.4	61.8	67.1	60.7	65.2	69.3	63.6	67.3	62.1	65.5	10.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	88.0	67.1	62.0	83.1	65.3	60.6	79.0	64.1	30.2	69.3	29.0	67.5	28.0	65.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.7	25.6	22.4	DB	-25.0	78.6	4.7	2.8	-28.3	80.6	-31.1	82.3	-33.7	83.8	-37.1	85.9	(4)
(5)			WB	-25.3	69.1	4.5	1.6	-28.6	70.3	-31.3	71.2	-33.8	72.1	-37.1	73.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.8	9.9	11.4	20.3	31.7	42.5	52.4	59.6	59.1	51.1	41.5	30.8	17.8		
	DBStd	18.92	11.84	11.38	9.96	6.16	5.20	5.29	4.08	4.50	5.53	5.71	8.13	10.75		
	HDD50	5959	1242	1082	920	550	238	34	0	2	50	267	577	997		
	HDD65	10665	1707	1502	1385	1000	697	379	174	188	416	728	1027	1462		
	CDD50	781	0	0	0	0	6	106	297	283	85	4	0	0		
	CDD65	12	0	0	0	0	0	1	6	5	0	0	0	0		
	CDH74	29	0	0	0	0	0	2	19	8	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	10.5	11.7	11.5	11.8	11.0	9.7	9.8	8.8	8.6	9.7	10.4	11.0	11.6		
	PrecAvg	46.5	3.7	3.0	3.5	3.3	3.7	3.7	3.6	4.1	3.9	4.4	4.9	4.7		
(15) Precipitation	PrecMax	56.1	6.9	7.1	6.8	6.1	9.3	6.9	6.9	6.6	7.3	9.8	7.8	9.4		
	PrecMin	30.8	1.7	0.7	1.2	1.0	1.0	1.5	0.7	1.7	1.5	1.2	2.3	2.8		
	PrecStd	6.1	1.3	1.4	1.5	1.4	1.8	1.5	1.6	1.3	1.4	2.0	1.5	1.7		
	PrecStd	6.1	1.3	1.4	1.5	1.4	1.8	1.5	1.6	1.3	1.4	2.0	1.5	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	39.2	41.3	50.1	63.8	72.8	76.4	74.8	69.1	58.9	50.6	40.8		
		MCWB	37.5	37.9	35.7	42.2	52.8	61.5	66.0	65.5	60.5	53.7	49.5	39.5		
	2%	DB	35.0	35.7	37.8	45.4	58.3	67.9	72.5	71.6	64.7	55.1	47.2	38.2		
		MCWB	34.4	34.3	34.3	39.6	48.9	58.3	63.4	63.3	59.1	51.5	46.0	36.9		
	5%	DB	32.4	32.7	35.5	42.2	54.8	64.5	69.7	69.1	62.1	52.6	44.8	35.9		
		MCWB	31.8	31.4	32.7	37.8	47.1	56.5	61.9	62.5	57.5	49.5	43.3	34.6		
	10%	DB	29.1	28.9	33.5	39.5	51.6	61.8	67.3	67.0	60.0	50.5	42.2	33.6		
		MCWB	28.0	27.4	31.3	35.9	45.5	54.8	60.9	61.5	56.2	47.7	40.5	32.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	38.3	37.4	43.6	54.6	63.6	68.4	67.7	63.5	55.4	50.1	40.1		
		MCDB	38.4	38.9	39.4	47.5	61.1	69.9	74.4	72.1	66.1	57.2	50.3	40.5		
	2%	WB	34.3	34.5	35.3	40.7	50.8	60.2	65.5	65.3	60.9	52.4	46.4	37.2		
		MCDB	34.8	35.3	37.3	44.3	56.0	65.6	69.7	69.2	63.2	54.0	46.9	37.8		
	5%	WB	31.9	31.4	33.2	38.5	48.3	57.9	63.6	63.9	58.9	50.3	43.7	34.8		
		MCDB	32.4	32.5	34.9	41.3	53.2	62.7	67.5	67.4	61.2	52.1	44.5	35.8		
	10%	WB	28.0	27.5	31.5	36.6	46.2	55.7	62.1	62.5	56.9	48.2	40.8	32.3		
		MCDB	29.0	29.0	33.2	39.1	50.4	60.3	65.7	65.7	59.4	50.1	41.9	33.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	17.8	16.1	12.3	14.4	14.7	14.0	14.1	14.1	12.6	12.0	14.7		
		MCDBR	17.0	13.6	13.2	15.0	20.2	19.7	18.6	17.1	16.3	14.7	11.5	13.3		
		MCWBR	17.4	13.3	10.8	10.5	12.4	11.3	10.8	10.4	11.3	11.7	11.4	13.8		
	5% WB	MCDBR	16.9	13.7	11.8	12.9	17.5	16.5	15.2	14.1	13.5	12.8	11.1	13.5		
		MCWBR	17.2	13.5	10.6	9.9	11.8	10.8	10.0	9.6	10.8	11.3	11.7	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.292	0.301	0.332	0.367	0.387	0.415	0.383	0.355	0.333	0.297	0.279		
	taud		2.367	2.323	2.378	2.409	2.405	2.369	2.341	2.433	2.481	2.502	2.512	2.400		
	Ebn at Noon		243	269	289	290	283	277	266	270	267	253	238	226		
	Edh at Noon		20	27	31	34	35	37	37	33	28	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		326	590	1011	1398	1626	1720	1646	1433	1083	672	346	251		
	RadStd		35	56	98	139	110	168	153	73	68	46	32	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+3.15	+1.19	+0.74	+0.74	N/A	N/A	+67	+5
Regional (7 neighbors)	N/A	N/A	+2.54	N/A	+0.68	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page